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AHA SCIENTIFIC STATEMENT

Coronary Atherosclerosis in Patients With Cancer and Survivors: A Scientific Statement From the American Heart Association

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ABSTRACT: There is an emerging convergence between atherosclerotic cardiovascular disease and cancer, driven by shared risk factors and overlapping pathophysiologic mechanisms. Traditional factors, such as smoking, aging, obesity, hypertension, and diabetes, alongside novel markers, such as clonal hematopoiesis of indeterminate potential, not only predispose individuals to both malignancies and coronary atherosclerosis but also amplify the risk of cardiotoxicity from cancer therapies. Inflammatory processes play a central role in atherogenesis, a process further accelerated by oncologic treatments—including chemotherapy (eg, anthracyclines, 5-fluorouracil), targeted and hormone therapies (eg, tyrosine kinase inhibitors, androgen deprivation, aromatase inhibitors), immune checkpoint inhibitors, and radiation therapy (RT)—that contribute to endothelial dysfunction and plaque instability. This scientific statement synthesizes the evidence on the interplay between cancer and coronary atherosclerosis, highlighting advances in noninvasive imaging modalities (ie, cardiac CT, nuclear imaging, cardiac magnetic resonance, echocardiography) for early detection, risk stratification, and surveillance of coronary artery disease in oncologic populations, and examines the role of invasive imaging techniques in guiding revascularization decisions. Given the elevated bleeding and thrombotic risks in these patients, individualized management of post–percutaneous coronary intervention medications and abbreviated dual antiplatelet therapy regimens is emphasized. This scientific statement also addresses knowledge gaps and reinforces the need for more evidence to improve risk stratification for atherosclerotic cardiovascular disease in patients with cancer. The shared pathobiology between coronary atherosclerosis and cancer necessitates an integrated, multidisciplinary approach to screening, diagnosis, and management.

Key Words: AHA Scientific Statements ■ cardiac imaging techniques ■ cardio-oncology ■ coronary artery disease ■ percutaneous coronary intervention ■ plaque, atherosclerotic

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There is an intertwined relationship between risk factors for atherosclerotic cardiovascular disease (ASCVD) and risk factors for cancer.[1] Smoking is the strongest link between ASCVD and cancer; aging, obesity, hypertension, and diabetes have been linked to increased rates of malignancies as well.[1] In addition, oncologic treatment modalities—including RT where the heart is in the treatment field, targeted therapies, immune checkpoint inhibitors (ICIs), and anthracyclines—have been linked to the development or progression of atherosclerosis.[2–5]

These relationships are of utmost importance not only because of the comorbid implications of coexisting cardiovascular disease (CVD) and cancer, but also because cardiovascular risk factors increase the risk of cardiotoxicity from cancer therapy.[6] These close relationships may explain the high CVD burden in patients with cancer and the need for aggressive cardiovascular prevention in these patients. It has been well-established that patients with cancer have a higher risk of complications from percutaneous coronary intervention (PCI).[7] Ethnic and racial disparities also exist for the development of CVD and cancer.[8,9] Inequalities in access to health care, environmental pollution exposure, food deserts, and other socioeconomic determinants of health may increase risk of morbidity and mortality in both cancer and CVD.

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Clonal hematopoiesis of indeterminate potential (CHIP) is a more recently established risk factor strongly associated with cancer and CVD. CHIP, in which proinflammatory clones of white blood cells originate from the bone marrow, is associated with an increased burden of coronary atherosclerosis, confers a 4-fold risk for CVD, and is linked (to a lesser degree) to an increased risk of hematologic malignancies.[10] CHIP results from sequence variations in *DNMT3A*, *TET2*, and *ASXL1*.[11] CRP (C-reactive protein), an acute phase reactant and marker of inflammation, has been found to identify increased ASCVD risk.[12]

INFLAMMATION AND ATHEROSCLEROSIS

Atherosclerosis is a chronic plaque-forming inflammatory disease of the vasculature involving multiple cell types, including the intimal endothelial cells (ECs) lining the vessel wall, surrounding medial smooth muscle cells, and cells of the innate and adaptive immune systems.[13] Plaque formation begins when ECs become damaged, typically because of risk factors including hypertension, diabetes, or dyslipidemia. EC damage leads to induction of surface cell adhesion molecules that recruit leukocytes into the vessel intima. Myeloid cells in the vessel wall engulf oxidized low-density lipoprotein, become lipid-laden foam cells, and may undergo cell death, thereby contributing to an unstable necrotic core. Medial smooth muscle cells dedifferentiate and migrate into the plaque, forming a plaque-stabilizing fibrous cap; others take on proinflammatory or procalcific phenotypes, further contributing to plaque instability. Proinflammatory cytokines released by plaque cells further recruit adaptive immune cells, including T cells, that produce proteases that degrade the protective fibrous cap.[14] Plaque erosion or rupture of unstable plaques exposes the prothrombogenic core, activating platelets and the coagulation system, resulting in arterial thrombosis, the most common cause of acute myocardial infarction (MI) and ischemic stroke.[15] In patients with cancer, in addition to shared atherosclerotic risk factors (ie, smoking, advanced age, CHIP), atherogenesis may be accelerated by EC damage from cancer drugs and RT, cancer treatments that exacerbate traditional risk factors, immune system activation by immunotherapies, and prothrombotic effects of cancer itself on platelets and coagulation proteins (Figure 1).

CORONARY ATHEROSCLEROSIS RELATED TO CANCER THERAPEUTICS

Chemotherapy

Chemotherapy is widely used in both hematologic and solid malignancies, and leads to improved clinical outcomes. However, many chemotherapeutic drugs are associated with cardiotoxicity (eg, left ventricular dysfunction), which can lead to early termination of curative or life-sustaining therapy. Less commonly studied is the impact of chemotherapy on ASCVD. Anthracyclines may cause endothelial dysfunction, which is an early event in the development of atherosclerosis (see Table 1[16–50] for a brief summary of cancer therapeutics and their association with coronary atherosclerosis).[16] Duncan et al[17] demonstrated increased rates of dyslipidemia, a known risk factor for atherosclerosis, in pediatric patients with hematologic cancers treated with anthracyclines, and Kalábová et al[18] demonstrated progression of atherosclerosis in patients with breast cancer treated with anthracyclines. Dong et al[19] observed higher rates of hyperlipidemia, elevated blood glucose, uric acid, and coronary artery disease (CAD) in patients with newly diagnosed breast cancer treated with anthracycline containing regimens. Patients with B-cell lymphoma treated with R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisolone) had significant increases in carotid intima-media thickness and plaque scores, increasing their risk of ASCVD.[20] 5-Fluorouracil, which is commonly used in colon, rectal, gastric, or breast cancer treatment, can cause coronary vasospasm and

Table 1. Relationship of Cancer Therapeutics to Coronary Atherosclerosis

Treatment	Coronary atherosclerosis risk association	References
Anthracyclines	May lead to endothelial dysfunction (early atherosclerosis) and dyslipidemia	16–20
5-Fluorouracil	May lead to coronary vasospasm and procoagulant state; unclear direct role in atherosclerosis	21
Tyrosine kinase inhibitors	Associated with cardiovascular toxicities, including coronary artery disease and arteriosclerosis Higher incidence of coronary artery disease, acute coronary syndrome, and ischemic stroke	22,23
Protease inhibitors	Increase risk of myocardial infarction, coronary artery disease, and hypertension	24,25
Androgen deprivation therapy	May augment risk by increasing visceral adiposity, hyperlipidemia, insulin resistance, and endothelial dysfunction	26,27
Aromatase inhibitors	Associated with higher risk of MI compared with tamoxifen	28
Immune checkpoint inhibitors	Accelerate atherosclerosis progression and increase noncalcified plaque volume and risk of atherosclerotic cardiovascular disease events	4,29–34
Therapeutic radiation	Generates free radicals and a proinflammatory state, leading to plaque rupture, thrombosis, and accelerated atherosclerosis	2,35–50

MI indicates myocardial infarction.

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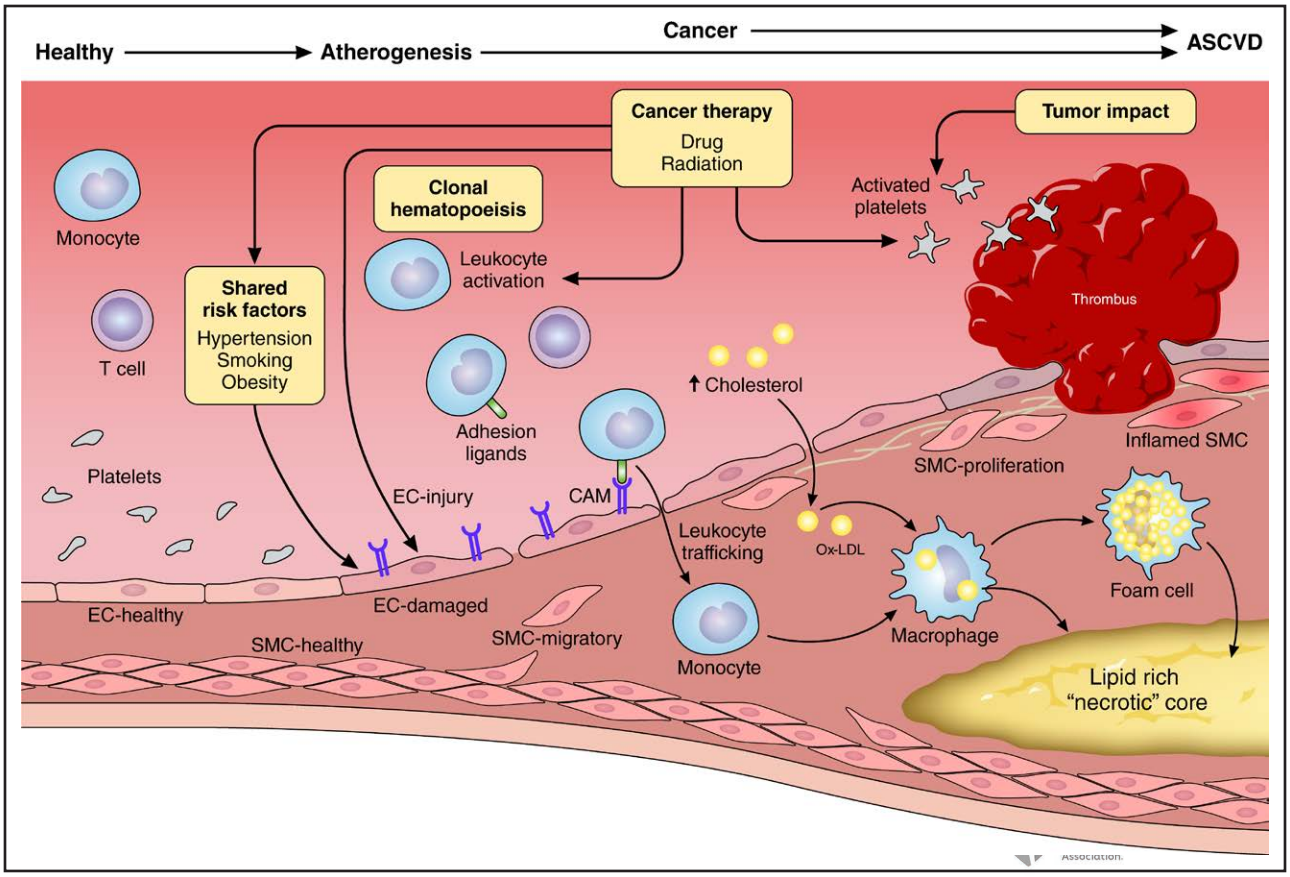

Figure 1. Coronary atherosclerosis–inducing pathways from shared risk factors, cardiotoxic-related factors, and malignancy-specific factors in patients with cancer and cancer survivors.
ASCVD indicates atherosclerotic cardiovascular disease; CAM, cell adhesion molecule; EC, endothelial cell; Ox-LDL, oxidized low-density lipoprotein; and SMC, smooth muscle cell.

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activation of a procoagulant state; however, the influence of 5-fluorouracil on the development of coronary atherosclerosis remains unclear.[21]

Targeted Therapies and Hormone Therapies

Targeted therapies have revolutionized the approach to cancer treatment but are associated with an increased risk of cardiotoxicity. Tyrosine kinase inhibitors (TKIs), which are used in the treatment of chronic myelogenous leukemia, renal cancer, and non-small cell lung cancer, are associated with several cardiovascular toxicities, including arrythmias, CAD, cardiomyopathy, and acute MI.[22] A higher risk of heart failure has been observed with imatinib (a first-generation TKI); nilotinib, dasatinib, and ponatinib (second- and third-generation ABL-TKIs) are associated with a higher incidence of CAD when compared with imatinib (13.4%) and a 3- to 4-fold increased risk of acute coronary syndrome, ischemic stroke, and acute limb ischemia.[23] The proteasome system is believed to play a role in the development and progression of atherosclerosis.[24] However, the effect of proteasome inhibitors on coronary atherosclerosis remains controversial. Bortezomib, carfilzomib, and ixazomib, which are commonly used in the treatment of multiple myeloma, have been associated with an increased risk of MI and CAD as well as hypertension and cardiovascular ischemia.[22] Carfilzomib is associated with a higher risk of an adverse cardiovascular event (18.1%) compared with control patients not receiving carfilzomib.[25]

Androgen deprivation therapy (ADT) is a cornerstone of prostate cancer treatment. Among the 3 available meta-analyses of observational trials, ADT was associated with an increased risk of cardiovascular events, cardiovascular death, and MI.[26] The increased risk of adverse cardiovascular outcomes from ADT is thought to be related to its role in promoting visceral adiposity, lipolysis inhibition to induce hyperlipidemia, insulin resistance, and endothelial dysfunction. This unfavorable cardiovascular risk profile may predispose ADT-treated patients with cancer to MI, stroke, and hypertension.[27] Aromatase inhibitors have been associated with higher risk of MI when compared with tamoxifen.[28]

Immunotherapies

Evolving evidence suggests that cancer immunotherapies may accelerate atherosclerosis and lead to an

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increased risk of ASCVD, such as acute MI, stroke, and peripheral arterial disease.[29,30] Drobni et al[4] reported a 3-fold higher risk for cardiovascular events (MI, coronary revascularization, and ischemic stroke) after starting an ICI in 2842 patients with cancer and 2842 matched controls. In the imaging substudy of 40 patients, the rate of progression of total aortic plaque volume was >3-fold higher with ICIs. In a subsequent imaging study consisting of 40 cases and 20 matched controls, Drobni et al[31] reported the annual progression rate for noncalcified plaque volume to be 7 times higher in the ICI group compared with the controls (11.2% versus 1.6% per year; P=0.001). However, the controls showed a greater progression in calcified plaque volume compared with the ICI group. Calabretta et al[32] reported increased vascular inflammation by evidence of fluorodeoxyglucose uptake in the large arteries after cancer immunotherapy with ICIs in 20 patients with melanoma. The specific activation of less-calcified lesions may suggest that ICIs could profoundly affect local innate immune cells, particularly in patients with underlying early and vulnerable atherosclerotic plaques. More recently, treatment with ICIs was associated with a higher rate of ASCVD events and a noticeable increase in coronary artery calcium (CAC) progression.[33] However, in an earlier large pharmacovigilance study using VigiBase, the incidence of MI and cerebral arterial ischemia in ICI-treated patients was found not to be higher than that reported in the full database.[34] Overall, data supporting a higher risk of ASCVD in patients with cancer receiving ICIs are increasing.

Radiation Therapy

RT that includes the heart in the treatment field increases the risk of atherosclerosis. Ionizing radiation results in free radical generation and a proinflammatory state, which contributes to atherosclerotic plaque rupture, platelet aggregation, thrombosis, and replacement of damaged coronary intima cells by myofibroblasts.[35] These changes accelerate the development of and consequences from atherosclerosis. Ostial coronary lesions are often seen in patients who underwent higher doses of RT from mantle radiation. Cardiac radiation exposure has been associated with an increased risk of CAD (Figure 2) in survivors of Hodgkin lymphoma,[36–38] breast cancer,[2,39,40] lung cancer,[41,42] esophageal cancer,[43,44] and pediatric cancers.[45] The risk of CAD increases with radiation dose to the heart,[2,36,45] and, specifically, with dose to the coronary arteries and left ventricle.[41,46–50]

To reduce toxicity, there has been a shift toward RT approaches that minimize cardiac radiation exposure.[51] Advanced RT techniques, daily image guidance, and respiratory motion management contribute to highly conformal dose delivery with sparing of adjacent normal tissues. Furthermore, radiation doses and targets, as well as patient selection for RT, have evolved to minimize the toxicity risk in survivors.[51]

PREVENTIVE RECOMMENDATIONS FROM GUIDELINES AND CONSENSUS DOCUMENTS IN CARDIO-ONCOLOGY

Given common shared risk factors between CVD and cancer, and an increased risk of CVD from some cancer therapies, societal recommendations address the prevention of atherosclerosis in patients with previous or current cancer.[52] The European Society of Cardiology (ESC) guidelines recommend a personalized and early approach, with the optimal timing for initiation of preventive strategies at the time of cancer diagnosis and before initiation of cancer treatment. In these guidelines, baseline assessment includes identification of a history of preexisting CVD as well as new, untreated, or uncontrolled medical and lifestyle cardiovascular risk factors. Class I recommendations for referral to a cardiologist include baseline abnormal troponin level, natriuretic peptide level, or transthoracic echocardiogram results in patients at risk for or with preexisting CVD as well as for any patient whose ECG results raise suspicion for CVD. CVD risk scores, such as the pooled cohort equation, can be used, but their application to patients with cancer has limitations, because traditional risk calculators underestimate the additional risk of cancer in this patient population.[53] Some additional cancer-specific and treatment-specific scores exist; however, they mainly focus on the risk for development of cardiomyopathy rather than atherosclerosis and ischemic events, or apply to only one type of cancer population.[54–56] Additional Society recommendations put forth specific cardiac imaging recommendations for identification of atherosclerosis and for CVD risk stratification, including assessment of coronary plaque on both noncardiac and cardiac computed tomography (CT), which are expanded upon later in this scientific statement.[57,58,59]

Lifestyle modifications mitigate not only cardiac risks but also cancer treatment side effects and cancer recurrence risk.[60] The American Heart Association's (AHA) Life's Essential 8 approach includes recommendations related to nutrition (ie, DASH [Dietary Approaches to Stop Hypertension]/Mediterranean diet), physical activity, smoking cessation, sleep health (ie, 7–9 hours/night), weight, cholesterol, blood sugar, and blood pressure management.[60] Adherence to Life's Essential 8 is significantly correlated with decreased incidence of elevated arterial stiffness in older adult cancer survivors.[60] In addition, a 10-point increase in Life's Essential 8 cardiac health score correlated with a deduction (0.43 m/s) in pulse wave velocity, an indicator of cardiovascular health.[60] Health care professional recommendations and patient education are facilitators to adherence of

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Figure 2. Illustrative case of multivessel coronary artery disease in a cancer survivor.
A 53-year-old man had a history of stage 4b diffuse large B-cell lymphoma in remission after 6 cycles of R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone) and methotrexate 3 years before clinical presentation. His total cholesterol level was 157 mg/dL and his low-density lipoprotein level was 111 mg/dL. He had no history of diabetes or essential hypertension. The pooled cohort equation result was 4.1%. Because of increased shortness of breath, the patient underwent pharmacologic nuclear stress single-photon emission computed tomography, which failed to show substantial reversible perfusion defects or evidence of transient ischemic dilatation (**A**), but poststress left ventricular ejection fraction was 43%. **B**, The calcium score from a previous coronary computed tomography examination showed considerable coronary calcium burden, with a total coronary artery calcification score of 2302. Coronary computed tomography revealed severe multivessel coronary atherosclerosis (**C**), which was confirmed by invasive coronary angiography (**D**). The patient underwent aortocoronary bypass surgery, which was successful. LAD indicates left anterior descending; OM, obtuse marginal; and RCA, right coronary artery.

modifiable behavior changes and minimization of lifelong cardiometabolic, comorbidity, and mortality risk.[61] However, evidence suggests that cancer survivors exhibit low adherence to lifestyle guidelines.[62] Patient awareness of their individual cardiac risk, patient readiness, self-efficacy, socioeconomic barriers, and mental health assessments are crucial to a holistic approach and pragmatic implementation.[63] Physical activity is widely recommended before, during, and after cancer treatment per the patient's functional ability, but multidisciplinary collaboration (eg, oncology, cardiac rehabilitation, nutrition) and knowledge of contraindications due to fluctuations in treatment-induced complications (eg, malnutrition, anemia, pain, increased fracture risk) is essential to balance the risk versus reward ratio and patient safety.[63–65] Further research is necessary to reveal the magnitude of the effect and establish dose–dependent causation between Life's Essential 8 lifestyle modifications and ASCVD in diverse cancer populations.[60]

Pharmacologic treatment for hypertension and lipid disorders for patients with cancer and cancer survivors

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should follow the 2023 chronic coronary disease guidelines, the 2018 cholesterol-lowering guidelines, and the 2019 prevention guidelines.[66–68] In patients with cancer and cancer survivors with known CAD, guideline-directed medical therapy should be assessed and optimized as secondary prevention. Limited evidence suggested that intensive blood pressure control (target systolic blood pressure <120 mm Hg) may provide additional benefits.[69]

NONINVASIVE CAD DIAGNOSIS AND SCREENING

Cardiac Imaging in Oncologic Patients

Noninvasive cardiac imaging is foundational to diagnosis of and screening for CAD in oncologic patients. We summarize the strengths, limitations, and suitability of different imaging modalities to assess CAD risk in patients with cancer and cancer survivors in Table 2 and discuss pertinent evidence and recommendations in the following.

Cardiac CT

CT is used in many oncologic disorders, and, more recently, has had a key role in cardiovascular care.[70] Cardiac CT is the most accurate noninvasive tool for detecting atherosclerotic plaque. Numerous large-scale studies in the general population have established the role of CAC scoring for risk assessment and of coronary CT angiography (CCTA) for evaluation of luminal patency in low- and intermediate-risk patients; for detailed plaque analysis; and, in some cases, for assessing resting left ventricular function.[71] The direct results of cardiac CT have also been translated to incidental findings of coronary calcifications in the heart (and all arterial beds) as part of chest and body imaging performed for cancer staging and for other indications. Several multisociety practice parameters have been published outlining the expanded role of cardiac CT in patients with cancer, which range from risk assessment based on oncologic-indicated scans to dedicated imaging in oncology patients with CVD risk factors or symptoms.[57,72,73] Most recently, CCTA has been applied to increasingly high-risk patients as innovations such as CT fractional flow reserve (FFR) and photon-counting CT have been used to mitigate low specificity for evaluation of intermediate and calcified coronary stenoses.[70,74] CAD-RADS 2.0 (Coronary Artery Disease Reporting and Data System) is a multisociety systematic approach that standardizes the coronary findings of CCTA and allows a semiquantitative visual approach to plaque quantification. Applying its use to CCTA interpretation, particularly in cardio-oncology patients, could allow optimized classification of ASCVD severity and identify progression and high-risk plaque in patients undergoing treatment with cardiotoxic agents.[75] As supported by American College of Cardiology (ACC)/AHA preventive guideline recommendations[67] and endorsed by the Society of Cardiovascular Computed Tomography/International Cardio-Oncology Society consensus document on cardiac CT in cardio-oncology,[57] if cardiovascular risk is uncertain, identification of CAC in noncontrast chest CT scans or CAC score scans should prompt initiation of treatment with statins. Appropriate preventive care should be implemented according to CAC score and CCTA findings.[57,67] Effective radiation dose reduction strategies include protocol optimization, such as using prospectively ECG-triggered axial scan protocols; lowering tube voltage; using novel scanners, such as wide detector array scanners (256–320 rows); and using second- and third-generation dual-source CT scanners with high-pitch spiral scan protocols enabling single-beat acquisition. In suitable patients, the acquisition protocols allowed by the newest CT hardware and software enable even lower absorbed doses (<1 mSv).[76]

Nuclear Cardiology

Nuclear myocardial perfusion imaging can reliably exclude or identify exercise-induced or pharmacologically induced coronary ischemia in patients with cancer.[77] Single-photon emission CT (SPECT) and positron emission tomography (PET) scans may also offer CAC score with attenuation correction CT, which may identify the presence and risk of ASCVD for preventive management, especially when ischemia is absent according to perfusion imaging. In patients with cancer, the perfusion defects do not correlate well with changes in regional wall motion or ejection fraction or underlying CAD, suggesting the importance of coronary microvascular dysfunction and coronary spasm in this population.[51,78]

PET MPI can offer superior assessment of myocardial ischemia due to its routine attenuation correction, higher spatial resolution, lower radiation exposure, greater diagnostic accuracy, shorter acquisition times, and improved prognostic utility. In addition, it is the reference noninvasive modality to diagnose coronary microvascular dysfunction.[79] PET MPI allows for measuring myocardial blood flow (MBF) and coronary flow reserve (CFR), which are central to coronary microvascular dysfunction diagnosis. For example, in 35 patients at a median interval of 4.3 years after chest RT, there were significant inverse correlations between mean radiation dose and global CFR and CFR in the left anterior descending artery territory.[80] In addition, low CFR was associated with the occurrence of major adverse cardiovascular events in cancer cohorts, highlighting its prognostic implication in these patients.[81,82] PET MPI with MBF quantification also offers superior accuracy compared with SPECT MPI in assessing balanced ischemia from left main or triple-vessel disease.[83] It allows for absolute measurement of MBF, providing more accurate information about ischemia than the relative assessments of SPECT, in which false-negatives can occur in the setting of balanced

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Table 2. Strengths, Limitations, and Suitability of Different Imaging Modalities to Assess Coronary Artery Disease Risk in Patients With Cancer and Cancer Survivors

Imaging modalities	Strengths	Limitations	Suitable patient population
Echocardiography	Easily accessible Low cost Comprehensive assessment of ventricular size and function and valvular function Regional wall motion abnormalities indicative of myocardial infarction No radiation	Dependent on operator experience, with subjectivity in interpretation and interobserver variability Difficult imaging quality when acoustic window is limited Unable to assess functional ischemia	First-line imaging test when patients report cardiac symptoms or signs Screening echocardiography 5 y after chest radiotherapy in asymptomatic patients
Stress echocardiography	Easily accessible Low cost Assessment of exercise tolerance, ventricular function, wall motion, and hemodynamics High specificity for ischemia assessment No radiation	Similar limitations as echocardiography Subjectivity in interpreting regional wall motion abnormalities Difficult imaging quality when acoustic window is limited Lower sensitivity in detecting CAD	Can be considered 5–10 y after chest radiotherapy in asymptomatic patients depending on ASCVD risk
Coronary CT	Can offer definitive evaluation of CAD with high negative predictive value Noninvasive high-resolution anatomic evaluation and direct visualization of CAD May identify and characterize high-risk plaque features Lower radiation exposure can be achieved	Iodinated contrast may limit use with advanced renal insufficiency Assessment of severity of coronary lesions may be limited by extensive calcification, which may be present in elderly patients or patients with previous radiation therapy Tachycardia, arrhythmias, or inability to perform breath hold may limit image quality Radiation exposure	Symptomatic patients with intermediate to high pretest likelihood of CAD, age <65 y, and less obstructive CAD suspected In asymptomatic patients, a lower threshold could be considered in patients 5 y after chest radiotherapy with other known atherogenic cancer treatment and high ASCVD risk
CAC based on routine chest CT	Obtained from routine care No added cost or radiation Can be used in risk stratification	Unable to precisely quantify CAC and the degree of coronary stenosis Not able to assess noncalcified plaque	Recommend screening CAC on all patients with existing chest CT or oncology PET/CT
SPECT/PET	PET offers better accuracy for ischemia assessment in people with obesity and can assess microvascular disease and balanced ischemia High sensitivity in assessing coronary ischemia Good for patients with known CAD, extensive calcification, or coronary interventions	Exposure to radiation SPECT may be susceptible to attenuation artifacts in people with obesity, may result in false-positives, and may miss balanced ischemia Availability of PET may be limited to larger medical centers	Symptomatic patients with intermediate to high risk, especially for patients >65 y, with substantial CAC burden, or with known CAD or coronary intervention
Stress CMR	Superior image quality and resolution, cross-sectional imaging Ability to characterize LV and pericardial tissue (for scarring, inflammation, or infiltrative processes) and masses No tissue attenuation artifacts compared with nuclear imaging	Cost and access Gadolinium cannot be administered with severe renal impairment Movement and arrhythmia may limit image quality Presence of metal devices or prostheses will limit the use of CMR	Symptomatic patients when other functional tests are inadequate

ASCVD indicates atherosclerotic cardiovascular disease; CAC, coronary artery calcium; CAD, coronary artery disease; CMR, cardiovascular magnetic resonance; CT, computed tomography; LV, left ventricle; PET, positron emission tomography; and SPECT, single-photon emission computed tomography.

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ischemia. This is particularly relevant to radiation-induced CAD, where the left main trunk, proximal left anterior descending, or right coronary arteries are often involved. Strategies for radiation dose reduction, such as weight-based radiotracer dose, ^{99m}Tc-labeled tracers, stress-first imaging, prone imaging, and novel scanners, are recommended for SPECT. PET MPI has the particular advantage of lower radiation dose due to shorter half-lives of radiotracers such as ^{82}Rb and ^{13}N-ammonia.[76]

Cardiac Magnetic Resonance

Cardiac magnetic resonance (CMR) is a comprehensive, robust, relatively fast, and cost-effective imaging test capable of assessing myocardial ischemia accurately and precisely.[84] The transmural extent of late gadolinium enhancement is a well-established marker of myocardial viability and can help identify myocardial territories that are likely to improve after revascularization.[85] Another unique ability of CMR is characterization of myocardial tissue

properties through the quantification of myocardial native T1, T2, T2*, and extracellular volume fraction.[86]

When combined with a vasodilator and first-pass contrast imaging, CMR can be used to detect myocardial ischemia due to obstructive coronary artery disease with a diagnostic accuracy that is superior to that of both stress echocardiography and stress SPECT[87,88] and similar to that of PET.[89] Among patients with stable angina, stress CMR was noninferior to invasive FFR with respect to major adverse cardiac events.[90] It is now possible to quantify MBF and calculate the myocardial perfusion reserve with CMR. Quantification of these measures has been shown to further improve the diagnostic accuracy of stress CMR,[91] better determine the myocardial ischemic burden,[92] and assess for MBF.[93] Both the AHA/ACC chest pain guideline[70] and the ESC chronic stable angina guideline[94] assign stress CMR a Class I indication for chest pain assessment.

During a routine CMR examination, other complications of CAD, such as mitral regurgitation, right ventricular failure, and left ventricular aneurysm and thrombus formation, may be identified. Because CMR can evaluate multiple facets of heart disease, it also has a valuable role for identifying specific causes of myocardial damage in patients presenting with MI who have nonobstructive coronary disease, such as stress-induced cardiomyopathy, myocarditis, or other acute conditions that are often encountered in patients with cancer.[95]

CMR is the gold standard noninvasive test in the diagnosis and risk stratification of ICI myocarditis.[96] The diagnosis of myocarditis relies on the modified Lake Louise criteria, which include nonischemic myocardial injury and myocardial edema as the primary criteria.[97] Based on current data, reliance on late gadolinium enhancement or T2-weighted short tau inversion recovery for diagnosis of ICI myocarditis may be insufficient.[98] Applying the modified Lake Louise criteria including T2-weighted images and late gadolinium enhancement along with T1 and T2 mapping and extracellular volume may offer improved diagnostic sensitivity. Among 136 patients diagnosed with ICI myocarditis who underwent CMR, 95% met nonischemic myocardial injury criteria and 53% met myocardial edema criteria with addition of T1 and T2 mapping. Native T1 values had excellent discriminatory value for subsequent major adverse cardiac events, with an area under the curve of 0.9.[99] In patients undergoing cardiotoxic therapies, CMR may be considered for patients with low normal or decreased left ventricular ejection fraction on echocardiogram, or discrepant left ventricular ejection fraction assessments on serial imaging, due to its higher spatial resolution and reproducibility for left ventricular ejection fraction assessment.[49] Parametric mapping, particularly using T1 and T2 mapping, and strain may represent a feasible and promising opportunity to detect early myocardial changes preceding overt cardiotoxicity.

Echocardiography

Echocardiography, a fundamental imaging technique in cardio-oncology, has an important role in cardiotoxicity monitoring, and provides information regarding baseline abnormalities that could suggest substantial CAD. Transthoracic 2-dimensional echocardiography can be helpful in identifying regional wall motion abnormalities at rest, especially with the assistance of ultrasound enhancing agents, suggesting underlying CAD.[100] Global longitudinal strain is used mainly as a measure to assess cardiotoxicity, but also may help identify ischemic or infarcted segments with segmental postsystolic shortening.[101] Both the wall motion score index at peak stress on stress echocardiography and the poststress left ventricular ejection fraction can serve as prognostic indicators.[102] The European Association of Cardiovascular Imaging/American Society of Echocardiography expert consensus for multimodality imaging evaluation of cardiovascular complications of radiotherapy in adults recommends the use of a functional noninvasive test, such as stress echocardiography, for CAD detection 5 to 10 years after exposure in high-risk asymptomatic patients.[103] In the European Society of Cardiology cardio-oncology guidelines, stress echocardiography is recommended as an optional modality to assess ischemia in symptomatic patients if there is clinical suspicion of CAD, especially before use of cancer therapies associated with vascular toxicity.[52] The guidelines assign a Class IIb recommendation for stress echocardiography or cardiopulmonary exercise testing in selected patients with exercise intolerance persisting 12 months after cancer treatment.[52] Echocardiography remains the primary imaging modality in the assessment of valvular heart disease secondary to chest radiotherapy from accelerated degenerative processes, including early valve retraction, thickening, and calcification, ultimately leading to stenosis or regurgitation. The aortic valve is most commonly affected, followed by the mitral valve. Echocardiography offers comprehensive assessment of structural and functional aspects of valvular heart disease, allowing for detailed visualization of valve morphology, Doppler interrogation, and qualitative and quantitative hemodynamic examination. For a more concise summary of the noninvasive imaging strategies to assess ASCVD risk in patients with cancer and cancer survivors, see Table 2.

Risk Stratification and Surveillance Strategies

Risk Assessment in Patients With Cancer and Cancer Survivors

Before patients receive cancer therapies that may increase ASCVD risk (ie, ADT [gonadotropin-releasing hormone agonists], ICI, nilotinib, ponatinib, RT, vascular endothelial growth factor inhibitors), risk stratification for ASCVD at baseline should be considered. Assessing preexisting CVD and cardiovascular risk factors through medical history, physical examination, and ECG

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are important steps to determine baseline ASCVD risk. Pooled cohort equations have been widely validated and are broadly used for the general population for predicting ASCVD risk. Although PCE are not the optimal risk prediction model for ASCVD risk in cancer populations based on previous reports, they are still the best-studied clinical risk score available. The Heart Failure Association/International Cardio-Oncology Society risk assessment tools mostly focus on cancer therapy–related cardiac dysfunction risk, and the new PREVENT score (Predicting Risk of Cardiovascular Disease Events) is yet to be studied and validated in cancer populations.[72,104] SCORE2 (Systematic Coronary Risk Evaluation) or SCORE2-OP (Systematic Coronary Risk Evaluation–Older Persons) is recommended by the ESC to assess baseline cardiovascular risk before chest RT.[52] Multiple guidelines recommended considering baseline echocardiography in patients with previous CVD before RT to a volume including the heart. It is also recommended to risk stratify chest RT–induced ASCVD risk based on mean heart dose. The ESC recommends considering patients with >15 to 25 Gy mean heart dose or ≥5 Gy mean heart dose + cumulative doxorubicin ≥100 mg/m^2 as high risk in general. Additional consideration can be given to some patients in whom a very small portion of the heart might be irradiated to a very high dose despite a low mean heart dose.[52]

Imaging Risk Stratification and Surveillance

Besides clinical factors and cancer treatment factors, standard serial CT imaging performed for cancer staging or treatment planning purposes provides an opportunity for baseline risk assessment and surveillance of the development of ASCVD, and this information can be used as directed by the Society of Cardiovascular Computed Tomography/International Cardio-Oncology Society consensus statement.[57] Despite limitations of CAC assessment in non–ECG-gated imaging, reasonable quantitative estimates have been described.[105] Early identification of CAC offers the opportunity for patient education, aggressive cardiovascular risk factor modification, and initiation of lipid-lowering medications, such as statins, which are generally underused in patients with cancer. If the CACS value is 0, the risk of dying from CAD within 5 years of cancer diagnosis remains below the mortality risk from cancer itself; conversely, if the CACS value is >300, the 5-year CAD mortality risk may exceed the cancer-related mortality risk.[106] The ACC Cardio-Oncology and Imaging Councils recommended early detection and treatment of subclinical atherosclerosis in cancer survivors and suggested consideration of dedicated CAC in high-risk patients and treatments to help guide preventative medical therapy, such as statins.[58] For patients who experience stable symptoms (ie, angina, limiting dyspnea) before, during, or after cancer therapy, functional imaging tests for myocardial ischemia—including stress echocardiography, stress CMR, or nuclear myocardial perfusion imaging—may be performed if there is an established history of CAD, especially considering the elevated risk of cardiovascular events with cancer therapy. In symptomatic patients with intermediate to high pretest likelihood of CAD, age <65 years, and less obstructive CAD suspected, CCTA is the preferred test.[70]

Multiple guidelines and expert consensus have suggested strategies to screen for ASCVD after cancer therapy posing ASCVD risk in asymptomatic patients. For example, the ACC/American Society of Echocardiography/European Association of Cardiovascular Imaging recommend screening echocardiography in the fifth year after chest radiotherapy, and noninvasive stress testing is recommended ≤5 years after treatment, with subsequent imaging then performed at 5-year intervals, with a preference for stress (dobutamine or exercise) echocardiography. In patients who underwent RT where the heart was in the treatment field and remain asymptomatic, a CAC score at 5- to 10-year intervals can be performed if there is no established history of CAD.[57] The ACC Cardio-Oncology and Imaging Councils suggested that after termination of radiotherapy, if the patient has a high risk of ASCVD, transthoracic echocardiography be performed after 5 years with or without stress echocardiography, and, if the patient is at low or intermediate risk, this should be done after 10 years in asymptomatic patients. A low threshold for a shorter follow-up interval and consideration of CCTA should be maintained when deemed appropriate.[58] The ESC recommended that screening for CAD should be considered in high-risk patients who have received chest RT to a treatment volume including the heart beginning at 5 years after RT using functional imaging or CCTA.[52] When new significant CAD (ie, intermediate stenosis typically between 40% and 90%) is detected on anatomic imaging, either FFR-CT or functional cardiac imaging should be considered to assess the need for revascularization, following the recommendations of the AHA/ACC chest pain guidelines.[70] Based on the available evidence, we summarize our recommendations on the use of different imaging modalities to assess CAD on the basis of patient characteristics in Table 2.

CORONARY INTERVENTIONS IN PATIENTS WITH CANCER

Invasive Imaging and Decision-Making

Coronary intervention should only be performed in patients with cancer or survivors with acute coronary syndromes or with symptomatic CAD despite optimal guideline-directed medical therapy.[107,108] Guideline-directed medical therapy should be assessed in all patients with known CAD and optimized when symptomatic,[68] based on the 2014 stable ischemic heart disease guidelines,[109] the 2018 cholesterol-lowering guidelines,[66] and the 2019 prevention guidelines.[67]

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There is a need for increased precision when considering revascularization in patients with cancer, because they have an increased bleeding and thrombotic risk.[110,111] Current low-risk revascularization options can improve a patient's resilience to disease and its treatments, but clinicians must also acknowledge that patients with cancer represent a prime substrate for type II MI, because anemia, hypotension, and tachycardia are common during cancer and its treatment. Practitioners should maintain awareness that after revascularization, patients with cancer typically can only tolerate an abbreviated dual antiplatelet therapy course to reduce bleeding risk and resume cancer therapy as soon as possible.

Complementary invasive procedures, such as FFR and instantaneous wave-free ratio, are essential tools for the functional assessment of intermediate and severe coronary lesions to limit procedures to those that are necessary in this higher-risk population. Using an FFR cutoff point of 0.75 to guide PCI translates into fewer interventions and can facilitate continuity of cancer care and reduced bleeding risk; patients with intermediate lesions (50%–70% stenosis as measured by quantitative coronary angiography) and FFR <0.75 are reclassified as severe and revascularized.[112] Large data set analysis in patients without cancer suggested that FFR-guided PCI is safe, and its use should increase: PCI does not significantly increase risk of inpatient mortality, complications, or length of stay.[113] Cancer and its treatments, despite their cytotoxic and antiproliferative effects, seemed not to delay stent healing or increase thrombotic risk in a study using contemporary drug-eluting stents (DES); however, a larger cohort is needed to confirm these findings.[114] An optical computed tomography–guided strategy allowed patients with cancer revascularized with DES and identified as low-risk for stent thrombosis to safely discontinue P2Y12 inhibition for cancer-related surgery or procedures.[115] Intravenous P2Y12 inhibitors have been used in patients with cancer needing to pause dual antiplatelet therapy for medical interventions.[116] Optical computed tomography can identify different lesion morphologies and personalize endovascular treatment.[117] Radiation to the mediastinum can result in CAD with a full spectrum of lesions, predominantly in ostial coronary segments. In patients with inflammatory coronary disease, such as Takayasu arteritis, intracoronary imaging can add diagnostic value due to its excellent correlation with histology.[118] Intracoronary imaging can also be used to ascertain the need for revascularization and can be used in patients with cancer who require revascularization for optimal stent expansion.[119]

Management of Post-PCI Medications in Patients With Cancer

As with all patients who undergo PCI, patients with cancer need individualized management. Special consideration must be given to factors such as coronary lesion location, cancer type and location, cancer therapy–related complications, and survivorship before considering intervention.[119] Bleeding complications pose dilemmas that complicate short- and long-term care. The risk of bleeding for patients with cancer is high and represents the most likely complication. Estimating the outcomes and bleeding risk is challenging given that patients with cancer were excluded in most trials. However, a report provides the framework of a modified PRECISE-DAPT score (Predicting Bleeding Complications in Patients Undergoing Stent Implantation and Subsequent Dual Antiplatelet Therapy) with cancer included as a risk factor that identifies patients with cancer who may carry a high bleeding risk.[120] Chemotherapy can theoretically delay stent endothelization, despite not being proven in vivo,[114] elevating thrombotic risk in an already prothrombotic population.[121] However, given the advent of newer generations of cardiac stents, imaging techniques, and several clinical trials including participants with high bleeding risk, reducing the duration of dual antiplatelet therapy to 1 to 3 months, as outlined in the literature, is recommended.[122,123] Choice of P2Y12 agent would be depend on their interactions with chemotherapy, platelet count, and other comorbidities, with some preference given to clopidogrel due to its potential lower bleeding risk.[123,124] If platelets are <50 000/μL, then consultation with a hematologist is advised.[125] Patients who require anticoagulation should ideally be treated with direct acting oral anticoagulation, assuming no interactions, and clopidogrel.[122,124] If statin therapy is initiated during the post-PCI period, selection of an agent with no chemotherapy interactions should occur.[124] Drug interactions related to statins and antiplatelet therapy should be considered in the setting of various cancer therapies.[124]

Complex Interventions

Because cardiovascular fitness drives overall survival in cardio-oncology, the need for coronary revascularization should be assessed before cancer therapy begins, and embedded in the patient's treatment plan.[126] Patients with CAD and cancer who undergo cardiac catheterization represent between 10% and 15% of total patients evaluated for coronary revascularization nationwide. The management of CVD in patients with active cancer presents unique challenges for interventional cardiology, because these patients often present with thrombocytopenia or a hypercoagulable state. Patients with cancer requiring PCI are included in several national PCI registries that have partially filled the gap left by the lack of randomized controlled trial evidence. Recent data emphasize the importance of aggressive management of cardiovascular risk factors and disease in improving cardiovascular health in this group of patients, regardless of cancer type or stage. PCI does not increase mortality risk or costs for patients with cancer; therefore, the presence

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or history of cancer should not preclude patients from receiving indicated cardiovascular care.[127] Cancer therapies are also associated with endothelial dysfunction and vascular toxicity, and hence can increase atherosclerosis event risk.[128,129]

Because patients with platelet counts >50 000/mm^3 had relatively normal thromboelastogram results and better cancer survival, a platelet count of 50 000/mm^3 can be considered normal for patients with cancer and CAD. Most patients with cancer with platelet counts <50 000/mm^3 have worse overall survival.[130]

Coronary angiography can be performed safely irrespective of the platelet count in patients with cancer with chronic thrombocytopenia. Unless ordered by oncology teams for other clinical indications, platelet transfusions are not favored before coronary intervention.[119] After consultation with primary oncology teams, aspirin can be administered in patients with platelet counts >10 000/mm^3. P2Y12 inhibition with clopidogrel may be used when platelet counts are 30 000–50 000/mm^3; for platelet counts <30 000/mm^3, consultation with the oncology team and a risk/benefit assessment is required.[131,132]

Limited data suggest similar survival and no increase in thrombotic risk in patients with cancer with CAD treated with DES or bare metal stents; therefore, a greater role for DES in these patients is to be anticipated.[133] Third-generation DES feature safer stent designs and biodegradable polymers, which may lead to better outcomes in patients with complex coronary anatomy.[128]

Early discontinuation of aspirin and P2Y12 inhibition should be anticipated due to increased bleeding risk from the underlying malignancy as well as unpredicted procedures during cancer treatment. An abbreviated dual antiplatelet therapy course <6 months after PCI appears not to be associated with stent thrombosis, but predicts worse cancer outcome.[134]

CONCLUSION, KNOWLEDGE GAPS, AND FUTURE DIRECTIONS

Emerging Diagnostic and Therapeutic Options

The detection and quantification of atherosclerotic plaque on CCTA have been traditionally based on visual assessment.[135,136] Quantitative measurements using manual or semiautomatic assessment are feasible and correlate with intravascular imaging; however, they are labor- and time-intensive.[137] More recently, advances in artificial intelligence, including deep learning and machine learning techniques, have enhanced the analysis of coronary atherosclerosis. These technologies enable automatic segmentation and quantification of coronary atherosclerotic plaque with high reproducibility in a fraction of the time required using previous methods. A study by Lin et al[138] showed that complete analysis of the coronary tree can be completed in <6 seconds with current technology. Automated coronary plaque measurements correlate closely with intravascular imaging.[138,139] Automated quantitative plaque measurements can classify patients into severity stages of coronary atherosclerosis and reliably predict the risk of future cardiovascular events.[140–142] Quantitative plaque assessment can also demonstrate compositional changes over time under preventive treatments (eg, statins) with plaque stabilization (eg, decrease in low CT attenuation plaque and increase in densely calcified plaque).[143] These new techniques provide an opportunity to guide the treatment of patients with cancer and cancer survivors who are at increased risk of coronary atherosclerosis. Those at highest risk may benefit from advanced preventive treatments (eg, PCSK9 inhibitors, bempedoic acid, inclisiran, colchicine). Further studies will clarify how best to use coronary plaque analysis to guide advanced therapies.

The detection and quantification of CAC from noncardiac CT scans provide a unique opportunity to harness existing data for enhancing risk assessment and instituting preventive measures in patients with cancer.[57,71] Artificial intelligence techniques can provide a reliable assessment of coronary calcium, coronary stenosis, and plaque features.[144–147] In the future, CAC analysis results could be incorporated into electronic medical records and prompt clinicians to consider preventive measures. This approach can lead to improved cardiovascular as well as cancer-related outcomes.[106,148,149]

Gaps and Future Directions

Atherosclerosis and cancer are diseases that often coexist, and complicate clinical care and lead to increased morbidity and mortality risk when both are present. Both are influenced by inflammation, genetics, environmental exposures, social determinants of health, and oncologic treatments. The lack of data on sex, racial, and ethnic differences in the impact of various cancer therapies on coronary atherosclerosis is a substantial knowledge gap requiring future research. This gap hinders understanding of how best to tailor treatments to minimize the risk of atherosclerosis in diverse populations. Global disparities in cardio-oncologic care and imaging access persist, affecting patients differently on the basis of their socioeconomic status, geographic location, and access to specialized care, which may affect timely detection and management of cardiotoxicity. Telemedicine and artificial intelligence have great potential to improve best practices implementation and alleviate disparities.

Patients with cancer need aggressive preventive care, and subclinical atherosclerosis should be detected in these patients, given their worse CVD outcomes and higher likelihood of receiving substandard preventive

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care.[150,151] Some groups (eg, patients with a history of RT where the heart is in the treatment field, those receiving ICIs, those receiving anti–vascular endothelial growth factor agents or some TKIs) may require further assessment if ASCVD risk factors are present even if they are asymptomatic. High-quality data on the efficacy and cost-effectiveness of surveillance strategies using different imaging modalities in the cardio-oncologic population relating to clinical outcomes are lacking. Whether statins and other preventive medications have similar efficacy in decreasing the risk of progression of atherosclerosis related to RT when compared with non–RT-related atherosclerosis is unknown, but it is reasonable to treat by extrapolation from robust data in noncancer cohorts.

CCTA is the best noninvasive tool to diagnose coronary plaque; adding artificial intelligence plaque quantification might help better assess risk of CVD in patients with cancer and cancer survivors. More research related to these diagnostic modalities in cardio-oncology will be important. Patients with cancer benefit from PCI when indicated, but are more vulnerable to vascular complications and bleeding when receiving PCI.[123] Whether other intrinsic factors, such as CHIP, may confer higher risk for both ASCVD and cancer is unknown, but its discovery sets a precedent for the possibility of finding similar markers. Although there are no randomized clinical data supporting that CCTA screening in asymptomatic patients improves outcomes, ongoing clinical trials and registries are evaluating this question (TRANSFORM [A Randomized Comparison of Stage-Based Care versus Risk Factor-Based Care for Prevention of Cardiovascular Events]).[152] Coronary plaque quantification technology using artificial intelligence has made these prognostic measures clinically applicable and practical.[140] With currently available technology, a lower threshold for using CCTA in patients with substantial risk factors or exposure to known atherogenic oncologic therapies, such as ICIs and RT, is reasonable.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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*Modest.

†Significant.

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